

Provakar Mondol

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RESEARCH INTEREST

- Deep Learning
- Medical Imaging
- Image Processing
- Remote Sensing
- Video Processing
- Bioinformatics

EDUCATION

- **Khulna University of Engineering and Technology (KUET)** Khulna, Bangladesh
M.Sc. in Electrical and Electronic Engineering; CGPA: 3.83/4.00 July 2021 - Ongoing
Thesis: *Optimized Deep Learning Models for Gastro-intestinal Image Classification and Automated Polyp Detection*
- **Khulna University of Engineering and Technology (KUET)** Khulna, Bangladesh
B.Sc. in Electrical and Electronic Engineering; CGPA: 3.52/4.00 March 2013 – May 2017
Last 2 years (81.25 credits) CGPA: 3.77
Thesis: *Design of A New Dry Electrode and Comparison of Performance with Biopac Wet Electrode*

RESEARCH EXPERIENCE

- **Gastrointestinal Image Classification and Automated Polyp Detection (2023-Present)**
 - **Multiclass GI image classification** with customized CNNs to support endoscopic inspection.
 - **Colorectal polyp segmentation** using U-Net + Watershed for improved clinical analysis.
 - **Real-time polyp detection** from colonoscopy videos using YOLO-based models.
 - **Published 2 IEEE international conference papers** on this work.
- **Framework for Diagnosis of Alzheimer's Disease in Data-Constrained Scenarios (2023-Present)**
 - **MRI-based Alzheimer's classification** using augmentation, SMOTE and WGAN-GP synthetic data generation to address class imbalances and scarcity of data.
 - **Transfer learning with three state-of-the-art CNNs** for performance benchmarking.
 - These works resulted in the publication of **2 IEEE international conference papers**.
- **Development of A Dry Electrode for Bio Signal Acquisition (2016-2017)**
 - **Developed a novel dry electrode** for biosignal acquisition (ECG, EEG, EMG).
 - **Benchmarked performance** against Biopac wet electrodes for validation.
- **Short Channel Effects Suppression in a Dual-Gate Gate-All-Around Si Nanowire Devices (2015-2016)**
 - **Designed a dual-gate GAA silicon nanowire nMOSFET** and compared short-channel parameters with conventional GAA.
 - Contributed to **2 IEEE international conference publications** based on this research.

PUBLICATIONS

- **P. Mondol**, P. Das, and K. K. Halder, "U-Net and Watershed Based Deep Learning Approach for Panoptic Segmentation of Colorectal Polyps in Colonoscopy," IEEE International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025. [\[Read Here\]](#)
- **P. Mondol**, P. Das, and K. K. Halder, "GINet: A Deep Learning Approach to Gastrointestinal Image Classification," IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025. [\[Read Here\]](#)
- P. Das, M. Islam, and **P. Mondol**, "Utilizing a Multi-Class CNN Model for Precise Diagnosis of Alzheimer's Disease," IEEE Global Conference on Cognitive Computing and Communication Technology (GC4T), Pune, India, 2025.
- P. Das, M. Islam, and **P. Mondol**, "A Comparative Analysis of Pre-trained Models for Identifying Alzheimer's Disease," IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025. [\[Read Here\]](#)
- M. W. Rony, **P. Mondol**, and H. R. Myler, "Sensitivity of a 10nm dual-gate GAA Si Nanowire

nMOSFET to Process Variation,” 19th International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2016. [\[Read Here\]](#)

- M. W. Rony, P. Bhowmik, H. R. Myler, and P. Mondol, “Short Channel Effects Suppression in a Dual-gate Gate-All-Around Si Nanowire Junctionless nMOSFET,” 9th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2016. [\[Read Here\]](#)

TECHNICAL & PERSONAL SKILLS

- **Programming Languages & Frameworks:** Python, C, C++, MATLAB, Keras, TensorFlow
- **Hardware & Circuit Design:** Arduino, Atmel AVR, Proteus
- **Graphics Programming and Industrial Applications:** Siemens SPPA T3000, MaxDNA, Valmet VMS, Honeywell PLC, Schneider PLC, Emerson PLC, ABB PLC

PROFESSIONAL EXPERIENCES

- **Bangladesh India Friendship Power Company Limited** Bagerhat, Bangladesh
Deputy Manager, Department of Control & Instrumentation December 2020 – Present
Responsibilities: Leading a team of 36 engineers in tasks like control loop tuning, GUI implementation, interlock checking, and network maintenance in a 1320 MW coal power plant. Completed procurement of goods and services worth approximately USD 2,100,000.00.
- **Daffodil International University (DIU)** Dhaka, Bangladesh
Lecturer, Department of EEE October 2017 – November 2020
Responsibilities: Taught courses and labs on Control Systems, Electrical Machine, Electronic Devices and Circuit Theory, Electronics, Random Signals and Processes, Power System Protection, Microprocessors and Interfacing. Supervised the thesis of 6 undergraduate students.
- **Prime University (DIU)** Dhaka, Bangladesh
Lecturer, Department of EEE August 2017 – September 2017
Responsibilities: Taught courses and labs on Control Systems, Electrical Machine, Electronic Devices and Circuit Theory to undergraduate students.

HONORS AND AWARDS (SELECTED)

- **Dean’s Award, KUET** – for outstanding academic performance (4th year).
- **Board Scholarship (Merit-13)** - Rajshahi Board, Higher Secondary Certificate Exam (2012).
- **Board Scholarship (Merit-41)** - Rajshahi Board, Secondary School Certificate Exam (2010).

SUPERVISION EXPERIENCES (SELECTED)

- **Mentor, Line-Follower Robot Competition (2016):** Supervised participants, enforced rules, and evaluated performance.
- **Core & Promotional Member, IUTF at KUET (2016):** Assisted in organizing and promoting a interuniversity tech competition platform.
- **Technical Operator, IEEE EICT Conference (2015):** Supported technical operations during an international academic conference.

STANDARD TEST SCORES

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| • TOEFL iBT - 106 | • GRE - 313 |
| Reading- 28, Listening- 29 | Quant- 167, Verbal- 146 |
| Speaking- 23, Writing- 26 | AWA- 4.0 |

REFERENCES

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| • Dr. Kalyan Kumar Halder
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